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# Longitudinal proxy pack benchmark generation for remaining useful life assessment of light electric vehicle battery packs in swapping systems

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**Abstract**

Battery swapping enables rapid energy replenishment for light electric vehicles (LEVs), but high-turnover pack circulation requires reliable pack-level health and remaining useful life (RUL) assessment. Publicly reusable longitudinal ageing datasets for complete LEV packs remain scarce. We develop a proxy-pack data-generation and learning framework that organizes independently tested cell-ageing records into fixed-member trajectories representing a 48 V-class 16-series configuration. Each trajectory retains 16 physical-cell identities over their common-cycle sequence. Cycle-level member states are aggregated into pack features for predicting a conservative minimum-member proxy RUL, with test barcodes isolated before combinatorial augmentation. Across ten barcode-level splits in each of three source cohorts, same-source tests with endpoints within the augmented training lifetime range achieved a median $R^2$ of 0.909 and a median Spearman correlation of 0.997. Tests beyond this range retained strong degradation ordering but showed larger absolute errors. A secondary mixed-source audit of 80 held-out trajectories achieved a median $R^2$ of 0.714 and a median Spearman correlation of 0.992. A matched composition-control analysis reduced the median $R^2$ by only 0.015 after balancing barcode reuse and limiting composition overlap, indicating limited sensitivity to training-composition sampling. A complementary physical audit using one fixed 12S3P pack confirmed cell-to-branch-to-pack electrical closure and revealed persistent voltage limitation and migration of the maximum-temperature location. Proxy-pack generation therefore provides a reproducible approach for constructing longitudinal member-aware benchmarks from cell-ageing records and evaluating pack-health and RUL models across varied compositions and lifetime conditions.

**Keywords** Light electric vehicle · Battery swapping · Remaining useful life · Proxy pack benchmark · Longitudinal data generation

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## Introduction

Light electric vehicles (LEVs), including electric two-wheelers and other low-voltage mobility platforms, support urban commuting, delivery services, and frequent short-distance travel. Compared with plug-in charging, battery swapping shortens replenishment time and turns energy provision into a managed process of pack circulation, centralized charging, and station-side screening[1–3]. In this setting, a swappable pack must be evaluated beyond state of charge. Its operational value also depends on state of health (SOH), remaining useful life (RUL), and suitability for future dispatch[4–6].

A central obstacle is data availability. Practical LEV swapping systems must balance low cost with limited sensing and communication under high pack turnover. Station-side or edge systems may have access only to estimated internal resistance, temperature extrema, charge time, capacity summaries, or other cycle-level descriptors. By contrast, pack-level lifetime labels are slow and expensive to obtain. Industrial pack records are also difficult to release because of commercial specifications, product identifiers, operating traces, and data-authorization constraints. Existing public vehicle-level and accelerated life-testing datasets provide useful references[7,8]. A recent pack-cycling dataset further provides member-cell voltages, branch currents, thermal measurements, and pack-level signals under vehicle-oriented dynamic profiles[9]. However, these datasets differ from low-voltage LEV swapping packs in their fields, protocols, configurations, application domains, or label definitions. Publicly reusable full-life LEV pack benchmarks that jointly provide member-cell information and pack-level health or RUL labels therefore remain scarce.

Cell-level SOH estimation and RUL prediction have advanced through experimental characterization, equivalent-circuit modelling, electrochemical modelling, data-driven learning, and physics-informed hybrid approaches[4–6]. Among public benchmarks, the MIT–Stanford fast-charging dataset is widely used for early-cycle lifetime prediction[10]. Subsequent studies have used this and related datasets to investigate data-efficient prognosis, semi-supervised learning, degradation-stage detection, uncertainty-aware prediction, and broader data-driven health estimation[11–13]. These studies show that cell-level cycling records contain strong degradation information before severe capacity fade becomes visible. Their data organization nevertheless remains cell-centric: one cell, one trajectory, and one lifetime label. This structure supports cell-level SOH and RUL prediction but does not directly define a pack-level learning target.

At pack level, health is not governed by an average cell alone. Cell-to-cell inconsistency, weakest-member limits, resistance and temperature dispersion, and state-of-charge imbalance can constrain usable energy, power capability, and ageing trajectories. New-pack assembly therefore generally seeks cells of the same specification and a common production or procurement cohort, followed by screening or matching of capacity, resistance, and other initial properties to reduce avoidable heterogeneity[14,15]. A common cohort nevertheless retains natural cell-to-cell variation. Existing work has investigated inconsistency-aware SOH estimation, pack-level SOH prediction under variable configurations, generalized series-pack prognostics, experiment-free inconsistency modelling, and cell-to-pack degradation analysis[16–21]. These studies establish that pack-level assessment requires explicit aggregation rules and representations of intra-pack heterogeneity. Many approaches, however, depend on physical pack experiments, richer sensing, predefined structures, or assumptions that are difficult to reproduce from public cell-level data alone.

A related question is whether member-aware health models remain reliable across source cohorts and operating domains. Battery health models are sensitive to differences among cells, ageing protocols, operating conditions, and target domains. Transfer learning, online correction, multi-domain adaptation, and limited-label calibration have therefore been proposed to improve robustness[22–26]. Few studies, however, examine the intermediate setting considered here: a member-aware learning object constructed from public cell records and evaluated under fixed-composition longitudinal holdout and cross-cohort stress tests.

To address this gap, we develop a longitudinal proxy-pack benchmark generation and learning framework for member-aware LEV battery assessment. The primary benchmark assigns 16 distinct cells from a common

source cohort to a fixed 48 V-class 16-series (16 S) composition and follows them over their exact common cycle sequence. Each trajectory preserves member identity, ageing order, and the natural variation in capacity, resistance, and degradation rate among nominally comparable cells. Controlled combinatorial reuse expands the range of training compositions, while held-out cell barcodes remain isolated during evaluation. Complementary cross-sectional and mixed-source experiments examine robustness to alternative construction settings and source-cohort variation. For the ideal all-series configuration, current continuity and voltage additivity provide the electrical basis for member aggregation. Because these circuit identities alone do not capture coupled pack operation, measurements from a physical 12S3P pack are used to audit electrical closure and evolving member-level limitations.

Three contributions distinguish this study from conventional cell-level RUL modelling. First, the framework converts public cell-ageing records into reproducible longitudinal proxy-pack benchmarks that retain both member histories and pack-level aggregates. Second, it evaluates member-aware RUL prediction under barcode-disjoint holdout and systematically examines endpoint coverage, composition sampling, feature and target choices, series-member count, and cross-cohort shift. Third, the measured 12S3P analysis provides a separate physical audit of cell-to-branch-to-pack aggregation and limiting-member dynamics. Although RUL prediction is the primary demonstration, the retained member-level structure also provides a basis for future studies of pack SOH, intra-pack dispersion, weak-member identification, abnormal compositions, and alternative maintenance-oriented targets. Together, these elements establish an auditable data-generation and evaluation framework for developing member-aware LEV battery-pack diagnostics.

## Problem formulation

Light electric vehicle (LEV) swapping stations circulate, charge, grade, dispatch, and retire complete battery packs. However, public longitudinal ageing datasets that jointly resolve pack-level outputs and individual member states remain scarce. Independently tested cell datasets, by contrast, contain substantially more complete ageing trajectories. New battery packs are generally assembled from nominally comparable cells selected from a common production or procurement cohort, often with additional screening for capacity, resistance, and other initial properties[14,15]. We therefore investigate whether the natural cell-to-cell variation within a public source cohort can be organized into fixed-member longitudinal proxy packs for member-aggregated remaining-life assessment. Here, a public source cohort is an experimental grouping and is not assumed to be a verified manufacturing lot.

This study focuses on compact all-series LEV packs containing tens of cells. A 48 V-class 16-series (16 S) configuration serves as the primary case, while 13–20 S configurations are examined in sensitivity analyses. Under the ideal all-series condition, the same current passes through every member cell, and the pack terminal voltage equals the sum of the member-cell voltages. These circuit identities provide a defined electrical basis for member aggregation without introducing current allocation among parallel branches. They do not reproduce synchronized pack trajectories, shared thermal fields, balancing actions, protection events, or electrically coupled degradation. Battery systems containing hundreds or thousands of cells would be better represented using hierarchical cell-to-module-to-pack constructions rather than unchanged flat member aggregation.

Let $s_{(c,t)}$ denote the observed state vector of cell c at cycle t. For the kth fixed composition, 16 distinct cell barcodes are assigned once and retained throughout their common observed cycle range. The corresponding proxy-pack snapshot is defined as:

$$\mathcal{P}_{(k,t)}^{(proxy)} = \{\boldsymbol{s}_{(c_{(k,1)},t)}, \boldsymbol{s}_{(c_{(k,2)},t)}, \cdots, \boldsymbol{s}_{(c_{(k,16)},t)}\} \tag{1}$$

The ordered sequence of these snapshots forms a fixed-member longitudinal proxy-pack trajectory. This construction preserves member identity, composition, and ageing order. Cross-sectional proxy instances serve a secondary role by recombining member observations to expand the range of cell identities and degradation-state

combinations used for augmentation and stress testing. Both constructions produce correlated learning instances rather than additional independent physical packs. The primary prediction target, stored as pack_rul_target, is the minimum-member proxy RUL:

$$RUL_{(c,t)} = EOL_c - t \tag{2}$$

$$y_{(k,t)}^{(RUL)} = min_{(i\in\{1,\ldots,16\})} RUL_{\left(c_{(k,i)},t\right)} \tag{3}$$

In Eqs. (2) and (3), $EOL_c$ denotes the observed end-of-life cycle of cell c, and t denotes its current cycle index. The minimum-member proxy RUL is a conservative algorithmic target motivated by weak-member limitation and cell inconsistency in series-connected systems[14,15,17–21,27,28]. It is not assumed to represent every physical pack-retirement mechanism. Lower-decile and mean-member targets are therefore evaluated as alternative definitions. The main supervised task is regression of $y_{(k,t)}^{(RUL)}$. Although RUL is the primary task, the retained trajectories allow other member-aware pack targets to be defined from the same benchmark structure.

The primary analysis evaluates RUL prediction from member-aggregated observations across complete fixed-member proxy-pack trajectories under barcode-disjoint same-source evaluation. Sensitivity analyses quantify the effects of lifetime-range coverage, composition sampling, feature selection, target definition, and series-member count. Mixed-source construction and cross-cohort testing examine robustness to compositional and distribution shifts. Complementing these proxy-based analyses, measurements from one fixed physical 12S3P pack provide a separate audit of electrical aggregation and limiting-member dynamics.

Within this formulation, "pack-level" denotes the level of representation, aggregation, and prediction rather than complete electrochemical or electrothermal pack simulation. Together, the proxy experiments and physical audit provide an intermediate evidence base between public cell-ageing records and longitudinal validation on multiple physical LEV packs.

## Methods

### Source cell cohorts and preprocessing

The source corpus comprises 138 physical cell barcodes and 108,995 usable cycle records extracted from the MIT–Stanford raw .mat files. The three public source cohorts contain 46, 48, and 44 cells, respectively, following the experimental grouping provided with the dataset. Each row represents one barcode at one valid cycle and records the cycle index, discharge capacity, internal resistance, temperature summaries, charge time, and derived health quantities. Records are retained if they occur after initialization, their cycle-summary fields are not all zero, their reference capacity and SOH are defined, and their RUL is non-negative. All evaluation partitions are defined at the physical-cell level using barcode identity. Within each evaluation fold, a barcode is assigned to only one data subset.

For cell $c$ at cycle $t$, the cell-level SOH is defined as the ratio between the current discharge capacity and a reference capacity estimated from the early valid cycles of the same cell:

$$\mathrm{SOH}_{c,t} = \frac{Q_{c,t}^{\mathrm{dis}}}{Q_c^{\mathrm{ref}}} \tag{4}$$

where $Q_{c,t}^{\mathrm{dis}}$ is the discharge capacity at cycle $t$, and $Q_c^{\mathrm{ref}}$ is the reference discharge capacity of cell $c$. In the current implementation, $Q_c^{\mathrm{ref}}$ is defined as the median discharge capacity over the first ten valid cycles of each cell.

## Proxy pack construction

### Same-source fixed-member longitudinal trajectories

Within each predefined barcode split, the primary protocol operates within one public source cohort. Sixteen distinct barcodes are assigned once to a fixed virtual 16 S test pack and excluded from the cohort-specific training pool. The same 16 members are retained over all cycle indices observed in common. The remaining 28–32 cells are used to generate 32 distinct fixed-member training compositions, each of which is followed over its common observed cycle range. No interpolation, carry-forward imputation, or lifetime-ratio alignment is applied. Observations are sampled at five-cycle intervals, and each trajectory receives equal total training weight. Training cells may recur across compositions, whereas test barcodes remain excluded from every training trajectory.

Reuse of training cells expands the range of member compositions available for model development. The dependence introduced by this augmentation is quantified using the number of unique compositions, pairwise barcode-overlap distributions, and barcode-use distributions. A matched composition-control analysis additionally balances barcode use and limits pairwise overlap to 12 members, as described in Sect. 3.5. Trajectories that share source-cell barcodes are treated as correlated augmentation samples.

### Supporting proxy constructions

Two supporting constructions test robustness beyond the primary same-source longitudinal setting. The mixed-source construction broadens the composition range while retaining member identity and common-cycle order. Cross-sectional constructions test whether predictive patterns persist when member observations are recombined at individual degradation stages, with and without lifetime-aligned pairing.

In the lifetime-aligned same-stage construction, member observations are combined at individual degradation stages rather than retained as fixed-member histories. Normalized cycle position, defined as cycle index divided by eventual complete cycle life, is divided into 100 bins. Starting from an anchor bin, the candidate search expands by at most ten bins. The nearest observation from each eligible barcode is retained, and 16 members are sampled from a candidate pool containing at most 64 observations. Eventual cell lifetime is therefore used for member pairing, although cycle-ratio features are excluded from model inputs.

The random-stage construction provides a lifetime-independent alternative. It samples 16 distinct barcodes and one observed cycle from each selected barcode without using eventual lifetime for pairing. Barcode identities remain isolated across the training, validation, and test partitions in both cross-sectional constructions. Because individual barcodes may recur across derived instances within a partition, these rows are treated as correlated augmentation samples.

## Features and targets

Member variables are converted into fixed-length statistical features at each longitudinal trajectory point and for each supporting cross-sectional instance. Each trajectory point or cross-sectional combination is treated as one proxy instance $k$. Let $u_{k,i,j}$ denote the $j$-th scalar feature of the $i$-th member in proxy pack $k$. Member-level variables are summarized using selected statistics from the mean, minimum, maximum, and standard deviation:

$$z_{k,j}^{(g)} = g\left(u_{k,1,j}, u_{k,2,j}, \ldots, u_{k,16,j}\right),\ g \in \text{mean, min, max, std} \tag{5}$$

The primary 27-feature leakage-controlled set comprises four discharge-capacity aggregates, four internal-resistance aggregates, four SOH aggregates, twelve temperature aggregates, and three charge-time aggregates. Mean statistics describe the central member tendency, minima and maxima retain limiting or extreme observations, and standard deviations characterize intra-pack dispersion. The 31-feature configuration additionally contains four cycle-ratio aggregates. Because cycle ratio depends on eventual complete lifetime, these columns are excluded

from all leakage-controlled model fits. The 23-feature core set additionally removes SOH, whereas the 19-feature low-sensing set removes both discharge capacity and SOH while retaining resistance, temperature, and charge-time descriptors.

For a fixed-member trajectory, the endpoint is defined by the earliest observed end of life among its members:

$$EOL_k^{\text{pack}} = \min_i \text{EOL}_{k,i} \tag{6}$$

At each common cycle t, the longitudinal target is the trajectory endpoint minus t and is therefore identical to the minimum member RUL at that cycle. No separate pack-SOH target is fitted in the proxy-RUL experiments. To test dependence on the chosen target, the cross-sectional sensitivity analysis also replaces the minimum with the lower decile or mean of member RUL values; the minimum remains the primary conservative proxy target $y_k^{\text{RUL}}$.

## Learning models and analytical tests

Two tabular regression models are evaluated on the MIT–Stanford-derived proxy-pack benchmark. Ridge regression is fitted after standardization using the training-data means and standard deviations. Its regularization parameter is selected from 0.1, 1, 10, and 100. The Random Forest search considers 200 or 400 trees, no maximum-depth constraint or maximum depths of 8 or 16, and minimum leaf sizes of 1 or 5. Hyperparameters are selected by validation-set MAE in the barcode-disjoint cross-sectional development experiment; test data are not used. The selected 27-feature Random Forest model uses 400 trees, a maximum depth of 16, and a minimum leaf size of 1. These values are fixed across all longitudinal experiments, repeated splits, and sensitivity analyses, with no split-specific retuning. Performance is evaluated using MAE (cycles), RMSE (cycles), Spearman rank correlation, and the coefficient of determination ($R^2$).

Grouped ablation and joint permutation analyses quantify the predictive information contributed by capacity state (discharge capacity and SOH), internal resistance, temperature, and charge time. In the permutation analysis, all columns within a feature group are permuted together using a common row permutation. This procedure preserves dependence among features within the group while breaking their association with the target. All feature comparisons use the same locked model and matched barcode partitions.

Member-count sensitivity is evaluated using 13, 14, 16, and 20 distinct cells under the random-stage all-series proxy definition. This analysis changes only the number of aggregated series members and does not represent parallel current sharing. Separately, fixed-member trajectories are used to track the cells with the lowest SOH, lowest discharge capacity, and highest internal resistance. These identities are also calculated using causal trailing 25-cycle medians, with at least five observations required, to reduce sensitivity to cycle-level fluctuations. Each estimate uses only information available at the current and preceding cycles.

## Evaluation and sensitivity analyses

The primary evaluation uses same-source fixed-member longitudinal trajectories. For each source cohort, 16 cell barcodes are assigned to one fixed test trajectory and excluded from the cells used for training. The remaining cells are recombined into 32 augmented training trajectories. This procedure is repeated across ten barcode-level splits for each of the three source cohorts, producing 30 cohort-by-split evaluations.

Endpoint-support analysis compares the endpoint of each held-out trajectory with the endpoints represented by its 32 augmented training trajectories. A held-out endpoint is classified as covered when it falls between the minimum and maximum training-trajectory endpoints. This analysis separates interpolation within the represented lifetime range from prediction beyond that range.

Composition-control sensitivity repeats the same 30 matched evaluations after balancing barcode use and limiting the overlap between any pair of 16 S training compositions to no more than 12 members. This comparison

isolates sensitivity to training-composition sampling. It does not treat trajectories sharing source cells as independent physical packs.

A secondary mixed-source evaluation examines a broader range of cell compositions. For each predefined regrouping seed, the 138 barcodes are shuffled and the first 128 are partitioned into eight mutually disjoint fixed 16-member trajectories. The remaining ten barcodes are unused in that regrouping. Each of the eight trajectories is held out once, while the other seven are used for training. Ten regroupings therefore produce 80 held-out trajectory evaluations.

The primary mixed-source result retains all selected cells, including the cell with the reported 148-cycle endpoint. A separate sensitivity analysis excludes only the ten held-out trajectories containing that cell. This comparison quantifies the influence of the short-lifetime endpoint without replacing the primary all-cell result.

The cross-sectional same-stage and random-stage stress tests use ten predefined barcode-level partitions. Each partition contains 97 training, 21 validation, and 20 test cell barcodes. The reported distributions and percentiles quantify sensitivity to barcode composition within the finite 138-cell corpus. They are not interpreted as confidence intervals for independent physical battery packs.

Strict leave-one-source-cohort-out evaluation is retained as an out-of-domain stress test rather than the primary assembly setting. Target adaptation is evaluated using barcode-disjoint calibration. Calibration and evaluation proxies are constructed from non-overlapping target-cell identities. Calibration budgets of 1, 5, 20, or 100 snapshots are generated from a separate group of 16 fully labelled target cells.

To characterize observable source-cohort shift at the physical-cell level, the first 100 cycles of each of the 138 barcodes were summarized using 14 early features: medians and interquartile ranges of discharge capacity, internal resistance, minimum, mean and maximum temperature, and charge time, together with reference discharge capacity and the number of valid early-cycle rows. SOH, feature slopes, and complete cycle life were excluded. A standardized multinomial logistic-regression classifier was evaluated using repeated stratified fivefold cross-validation with ten repeats and compared with a class-prior dummy classifier. Figure 5a reports mean balanced accuracy and the 2.5th–97.5th percentile range across the 50 validation folds. For each feature, source-cohort effect size was calculated as $\eta^2$, defined as the between-cohort sum of squares divided by the total sum of squares; the three largest $\eta^2$ values are highlighted in Fig. 5b. These quantities describe observable cohort and protocol signatures and are not interpreted as causal effects.

Statistical comparisons are summarized across matched barcode partitions rather than individual proxy rows because cell barcodes recur across derived instances. Model, feature, and construction comparisons are therefore reported descriptively across matched partitions. Figure 1 summarizes the analysis workflow and evidence hierarchy.

## Physical 12S3P electrical closure and member dynamics

A complementary physical analysis uses the public CC–CV/WLTP battery-pack dataset[9]. The proxy framework represents a pack through member states and their aggregate behaviour. This analysis therefore tests electrical closure across measured cell, branch, and pack signals and tracks limiting-member identities under coupled operation. The dataset records one fixed 12S3P lithium-ion pack over 286 complete and consecutive cycles, spanning cycles 125–410. These records comprise 217 WLTP cycles and 69 capacity-check cycles. Available measurements include 36 member-cell voltages, 72 top and bottom cell-surface temperatures, three branch currents, three branch voltages, total pack current, and two pack-voltage measurements. The pack is analysed in its native 12S3P configuration, providing a physical audit of member-aware aggregation rather than a direct evaluation of 16 S proxy-RUL prediction.

Electrical closure is evaluated at every timestamp using four classes of residual. These comprise total pack current minus the sum of the three branch currents, and each branch voltage minus the sum of its twelve cell voltages. They also include pack voltage minus the mean branch voltage and the difference between the two pack-voltage measurements. Pooled quantiles and graphical summaries use up to 1,000 regularly spaced timestamps

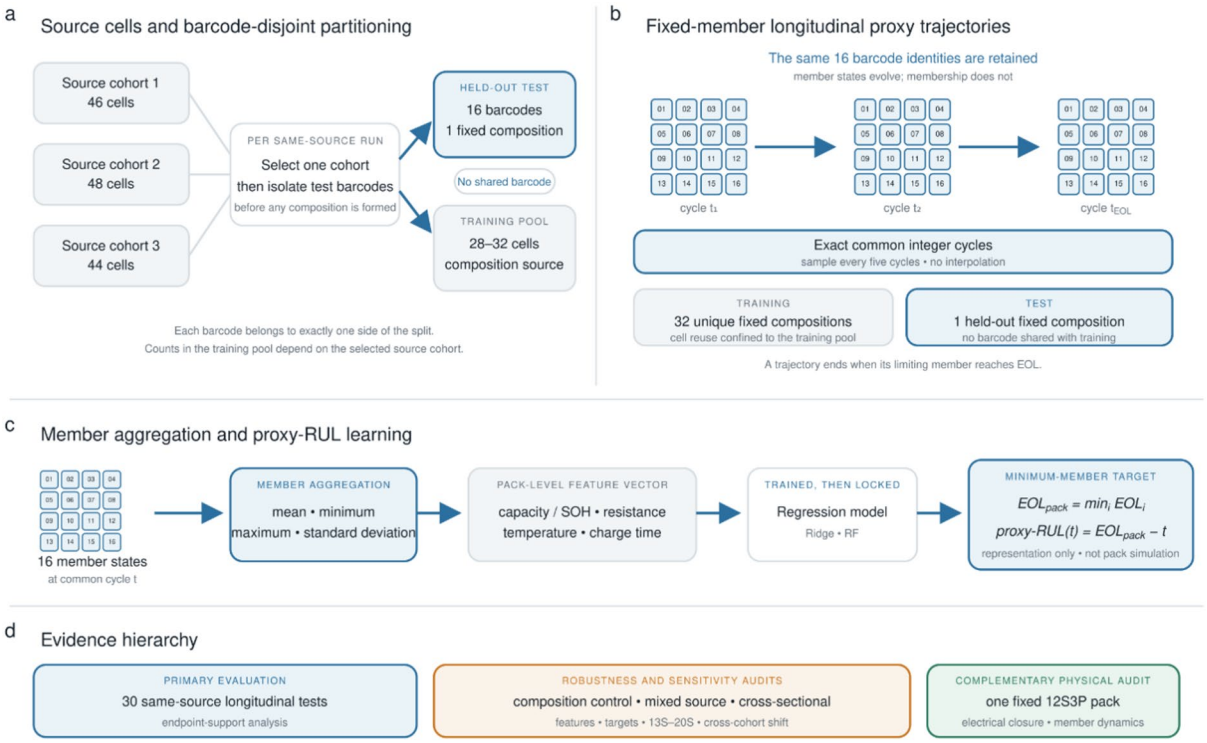



**Fig. 1** Proxy-pack benchmark construction and evidence hierarchy. (**a**) Source cohorts and barcode-disjoint partitioning. (**b**) Same-source fixed-member longitudinal trajectory construction. (**c**) Member aggregation and proxy-RUL learning. (**d**) Primary evaluation, robustness analyses and complementary 12S3P physical audit.

**Table 1** Same-source-cohort fixed-member longitudinal results across ten barcode splits per cohort.

| Source cohort | Feature set | Splits | Median MAE (cycles) | $R^2$ range (2.5th–97.5th) | Median Spearman | Median $R^2$ |
|---|---|---|---|---|---|---|
| 2017-05-12 | Full 27 | 10 | 24.05 | 0.760–0.991 | 0.999 | 0.971 |
| | Low-sensing 19 | 10 | 50.68 | 0.294–0.916 | 0.959 | 0.816 |
| 2017-06-30 | Full 27 | 10 | 86.09 | −9.501–0.853 | 0.975 | −2.552 |
| | Low-sensing 19 | 10 | 74.76 | −29.193–0.740 | 0.866 | −0.363 |
| 2018-04-12 | Full 27 | 10 | 76.86 | −0.065–0.934 | 0.999 | 0.716 |
| | Low-sensing 19 | 10 | 123.56 | −0.412–0.821 | 0.965 | 0.339 |

from each file. Per-cycle summaries are calculated from all available timestamps. Current-closure errors and branch-current spread are additionally summarized during active operation, defined by an absolute pack current above 0.2 A. Because parallel branches share a common terminal voltage but may carry unequal currents[29–31], branch-current imbalance is quantified using the mean and 95th percentile of the instantaneous current spread.

Member dynamics are evaluated during discharge, defined by a pack current below −0.2 A. Cell voltages outside 2–5 V and temperatures outside −20–100 °C are treated as invalid or sentinel values. These measurements are excluded before member identities are calculated.

At each discharge timestamp, the member with the minimum cell voltage and the sensor with the maximum temperature are identified. The dominant identity for each cycle is defined as the member or sensor occurring most frequently in the corresponding timestamp-level sequence. The dominant fraction and the number of distinct identities observed within each cycle are also recorded. Identity transitions between consecutive cycles are counted separately for the WLTP and capacity-check records.

Capacity-check capacity is calculated by integrating the measured pack current over elapsed time. Its association with cycle number and the consistency of the discharge-cutoff protocol are examined to assess its suitability

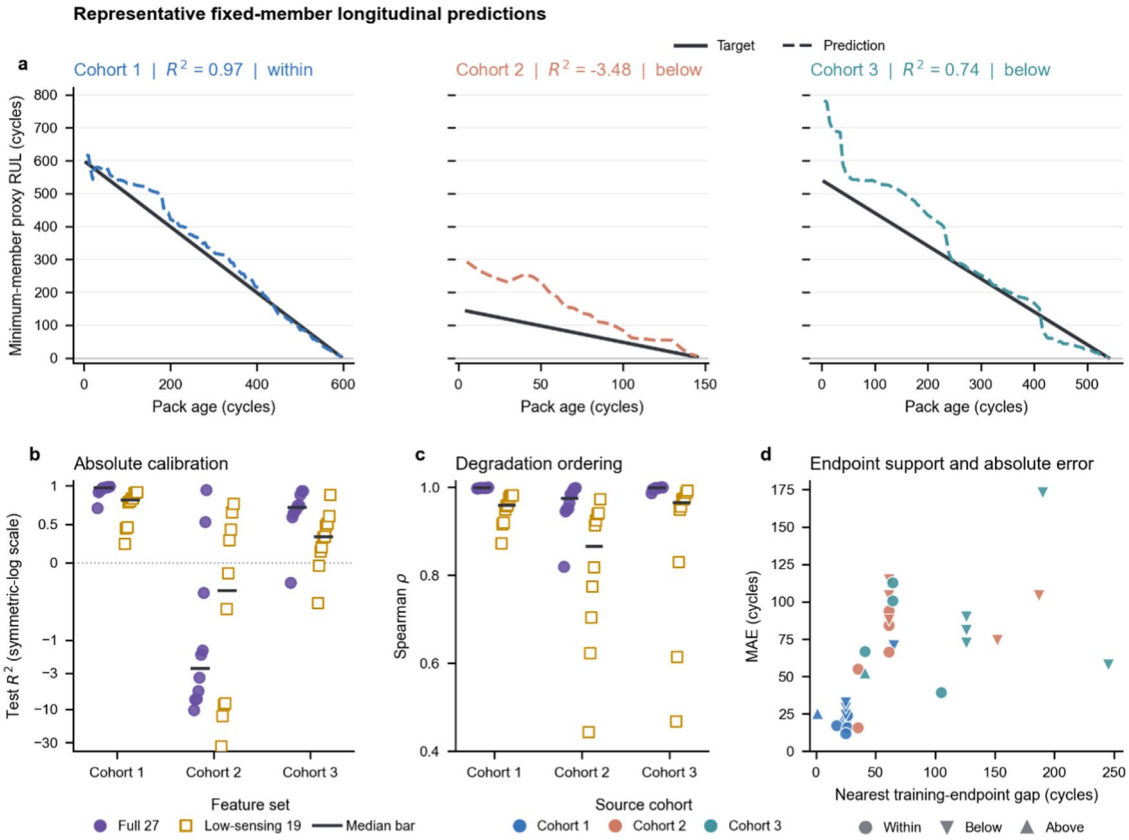



**Fig. 2** Same-source-cohort fixed-member longitudinal prediction for $y_k^{\mathrm{RUL}}$, showing representative trajectories, split-level absolute calibration, rank consistency, and endpoint-support effects.

**Table 2** Mixed-source fixed-member longitudinal prediction and endpoint-support audit.

| Evaluation subset | Trajectories | MAE (cycles) | Spearman | Median $R^2$ |
|---|---|---|---|---|
| All held-out trajectories | 80 | 57.34 | 0.992 | 0.714 |
| Endpoint within training range | 60 | 50.88 | 0.991 | 0.775 |
| Endpoint below training range | 10 | 154.30 | 0.992 | −14.002 |

as an external ageing indicator. Capacity is treated as a data-quality descriptor and is not used as a supervised target for the proxy-RUL model.

## Results

### Same-source fixed-member longitudinal RUL assessment

Same-source-cohort experiments defined the primary assembly-oriented setting: training and test members were drawn from the same public source cohort but remained barcode-disjoint. For the full 27-feature model, median $R^2$ values were 0.971, −2.552 and 0.716 for the 2017-05-12, 2017-06-30 and 2018-04-12 cohorts, respectively. The corresponding median Spearman correlations were 0.999, 0.975 and 0.999. The 19-feature low-sensing model achieved median $R^2$ values of 0.816, −0.363 and 0.339, with corresponding median Spearman correlations of 0.959, 0.866 and 0.965. Relative to the full model, performance declined in the first and third cohorts,

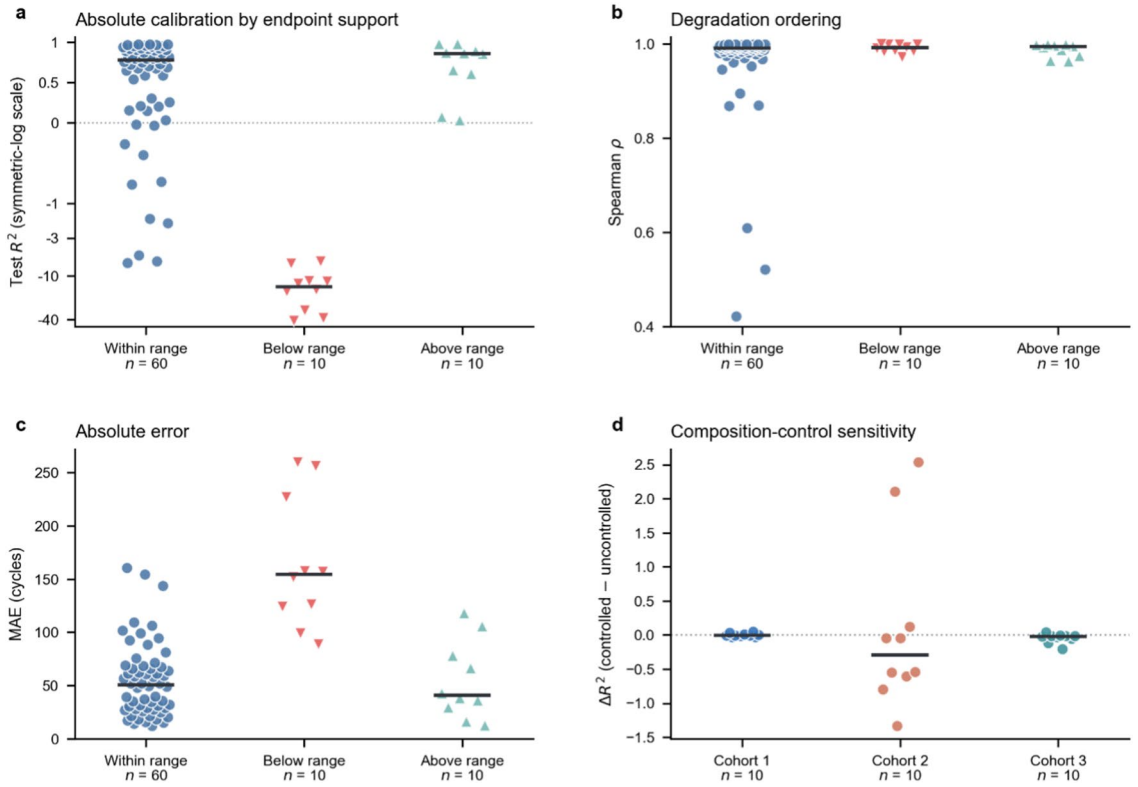



**Fig. 3** Lifetime-support and composition-control analyses. (**a**) Test R², (**b**) Spearman correlation, and (**c**) MAE for the 80 mixed-source held-out trajectories, stratified by whether the test endpoint lay within, below, or above the training-endpoint range. (**d**) Controlled-minus-original change in R² after balancing barcode reuse and limiting pairwise composition overlap across ten matched tests per source cohort. Horizontal bars indicate medians.

**Table 3** Repeated random-stage construction and sensitivity summaries.

| Construction | Splits | Median MAE | Median RMSE | Spearman | Median $R^2$ |
|---|---|---|---|---|---|
| Same-stage, 16 S | 10 | 27.81 | 35.39 | 0.985 | 0.932 |
| Random-stage, 16 S | 10 | 10.10 | 16.42 | 0.922 | 0.800 |
| Random-stage, 13 S | 10 | 12.39 | 20.14 | 0.928 | 0.787 |
| Random-stage, 20 S | 10 | 8.91 | 14.33 | 0.923 | 0.778 |

whereas absolute calibration remained poor under both feature sets in the second cohort. The low-sensing model also showed broader split-level R² distributions, particularly in the 2017-06-30 cohort.

## Lifetime support and mixed-source composition stress test

Retaining member identity over time converts the proxy object from a recombined row into a complete virtual-pack ageing trajectory. As a broader composition stress test, 128 pooled barcodes formed eight mutually disjoint 16-member packs per regrouping and each pack was held out once. Across 80 held-out trajectories, the full-feature Random Forest predicted the minimum-member proxy RUL with median MAE of 57.34 cycles, median RMSE of 70.21 cycles, median Spearman correlation of 0.992 and median R² of 0.714. Member-aggregated features therefore tracked degradation order consistently, while absolute endpoint prediction varied among held-out compositions.

The dominant calibration failure was lack of lifetime support. Among 60 trajectories whose endpoints lay within the training-pack range, median R² was 0.775 and median MAE was 50.88 cycles. The ten below-range

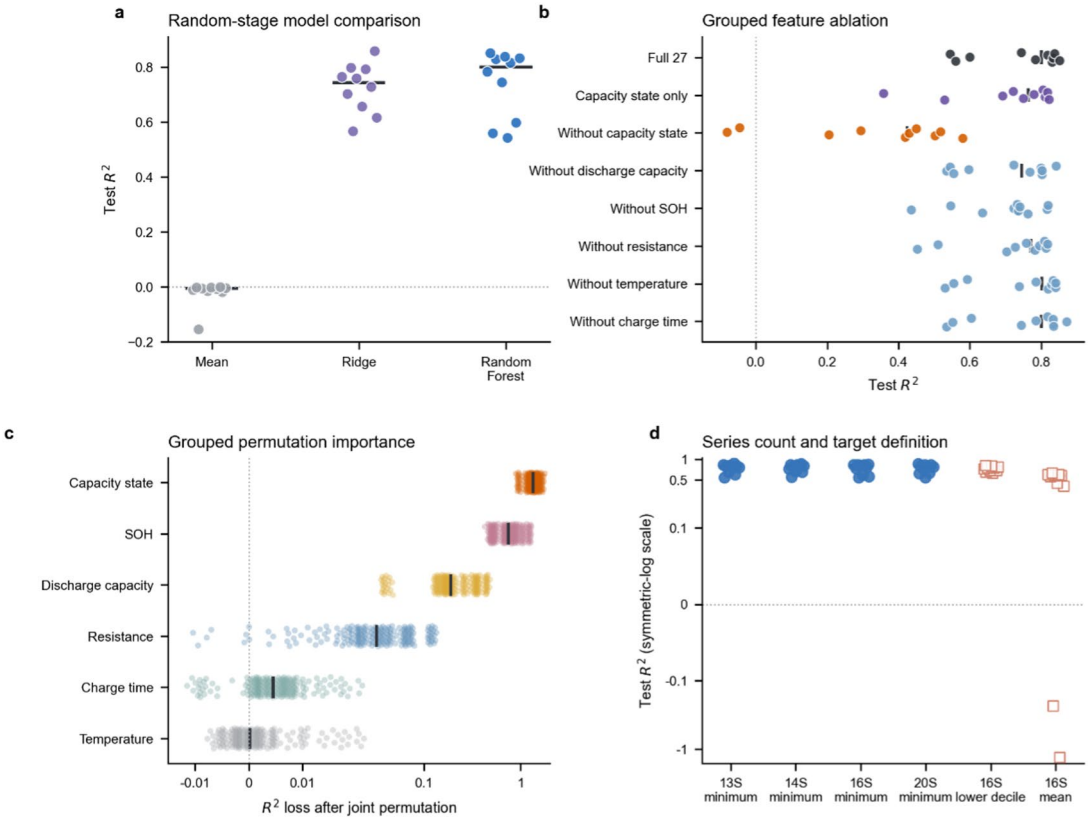



**Fig. 4** Random-stage model, feature, and proxy-definition sensitivity. (**a**) Test $R^2$ for the mean baseline, Ridge regression, and Random Forest across ten barcode-level splits. (**b**) Matched grouped feature ablation. (**c**) Grouped permutation importance measured by the loss in $R^2$ after joint permutation. (**d**) Sensitivity to series-member count and pack-level target definition. Horizontal bars indicate medians. Panel c uses a symmetric-logarithmic horizontal scale, and panel d uses a symmetric-logarithmic vertical scale.

**Table 4** Barcode-disjoint target-cohort calibration audit.

| Calibration | Snapshots | Mean MAE | Mean RMSE | Spearman | Mean $R^2$ | Target-cell cost |
|---|---|---|---|---|---|---|
| Source only | 0 | 142.45 | 180.46 | 0.930 | −1.748 | None |
| Target calibration | 1 | 140.39 | 177.85 | 0.932 | −1.732 | 16 labelled cells |
| Target calibration | 5 | 124.74 | 161.17 | 0.925 | −1.360 | 16 labelled cells |
| Target calibration | 20 | 95.53 | 123.77 | 0.926 | −0.439 | 16 labelled cells |
| Target calibration | 100 | 75.36 | 100.47 | 0.935 | 0.034 | 16 labelled cells |

Note: Each calibration condition uses a separate set of 16 target-cell barcodes with complete-life labels. Proxy snapshots are not independent target packs.

trajectories had median $R^2$ of −14.002 despite median Spearman correlation of 0.992; every such trajectory contained the unique 148-cycle short-life cell. Its 147 trainable records covered consecutive cycles 2–148, discharge capacity decreased smoothly from 1.072 to 0.878 Ah, and all records passed the predefined validity rules; 22 post-endpoint rows with negative RUL were already excluded. The cell was therefore retained as an extreme but valid short-lifetime case rather than removed according to model performance. When its ten held-out trajectories were omitted only as a sensitivity summary, median $R^2$ increased from 0.714 to 0.796 and median MAE decreased from 57.34 to 49.45 cycles. The tree model preserved within-trajectory ordering but could not extrapolate an endpoint below its training support.

Lifetime support also explained the variation within the primary same-source setting. Across the 14 same-source tests whose endpoints lay inside the augmented training-endpoint range, median $R^2$ was 0.909 and median

Spearman correlation was 0.997. The nearest endpoint gap correlated with MAE at Spearman $\rho = 0.722$ and with $R^2$ at $\rho = -0.647$. Drawing cells from one source cohort therefore reduces one domain shift, but it does not remove the need for the training trajectories to cover the target lifetime range.

## Combinatorial augmentation, member-reuse sensitivity and proxy analysis

We next examined whether the longitudinal result depended on how the augmented training compositions were sampled and which member-level information carried the predictive signal. The analyses in this section therefore progress from composition reuse to cross-sectional augmentation, grouped feature attribution and proxy-definition sensitivity.

The overlap-and-reuse sensitivity construction reduced dependence on repeatedly selected training cells without changing the central prediction finding. Across the three cohorts, the standard deviation of barcode reuse counts decreased from 2.78 to 2.98 in the original unique random compositions to 0.43–0.45 under balanced reuse, while all 32 training compositions per split remained unique and maximum pairwise overlap was limited to 12 members. Across the 30 matched tests, the median controlled-minus-original change was −0.0147 for $R^2$ and +0.0004 for Spearman correlation. The longitudinal result was therefore robust to substantially more even use of the available training cells.

The cross-sectional random-stage construction provides a stricter combinatorial stress test because eventual complete lifetime is not used to pair members. Across ten barcode-level splits, it achieved median $R^2$ of 0.800 (2.5th–97.5th percentile, 0.548–0.848) and median Spearman correlation of 0.922 (0.829–0.967). The submitted same-stage split reproduced $R^2 = 0.9362$ and Spearman = 0.9875, but repeated same-stage $R^2$ ranged from −2.279 to 0.974 and was strongly associated with test-target composition. The random-stage result is therefore used as the central cross-sectional estimate.

Feature attribution shows that the signal is concentrated rather than broadly multiphysical. The 27 fitted columns contain no cycle-life, cycle-ratio, RUL or target field. Nevertheless, discharge capacity and SOH are observable ageing proxies: removing these eight capacity-state features reduced median random-stage $R^2$ from 0.800 to 0.424, whereas the eight features alone achieved 0.764. Joint permutation caused a median $R^2$ loss of 1.352 for capacity state, compared with 0.032 for resistance, 0.005 for charge time and approximately zero for temperature. The model is consequently interpreted as capacity-state-dominated member aggregation.

Within the all-series proxy definition, changing the member count from 13 to 14, 16 or 20 produced similar median minimum-label $R^2$ values of 0.787, 0.789, 0.800 and 0.778. At 16 members, replacing the minimum target with the lower decile or mean reduced median $R^2$ to 0.743 and 0.577, and minimum-versus-mean target concordance was only 0.259. These results support limited robustness to series-member count within the defined proxy object while confirming that performance remains dependent on the chosen pack-level target.

## Cross-cohort domain-shift stress test

The three source cohorts were strongly separable using early observable cell summaries. The standardized multinomial classifier achieved mean balanced accuracy of 0.985 (2.5th–97.5th percentile, 0.903–1.000), compared with 0.333 for the class-prior dummy baseline (Fig. 5a). The largest univariate cohort effects were minimum-temperature median ($\eta^2 = 0.658$), charge-time interquartile range ($\eta^2 = 0.610$), and internal-resistance median ($\eta^2 = 0.592$; Fig. 5b). These signatures demonstrate observable cohort and protocol differences but do not distinguish manufacturing, cell-composition, or test-protocol causes.

Source-cohort shift remained severe when the model was required to extrapolate beyond the assembly-oriented same-source setting. In the barcode-disjoint audit, source-only mean $R^2$ was −1.748; calibration with 1, 5, 20 and 100 snapshots constructed from 16 separate labelled target cells yielded −1.732, −1.360, −0.439 and 0.034. The earlier 5–20% recovery was therefore an identity-overlapping, cohort-specific adjustment rather than evidence of

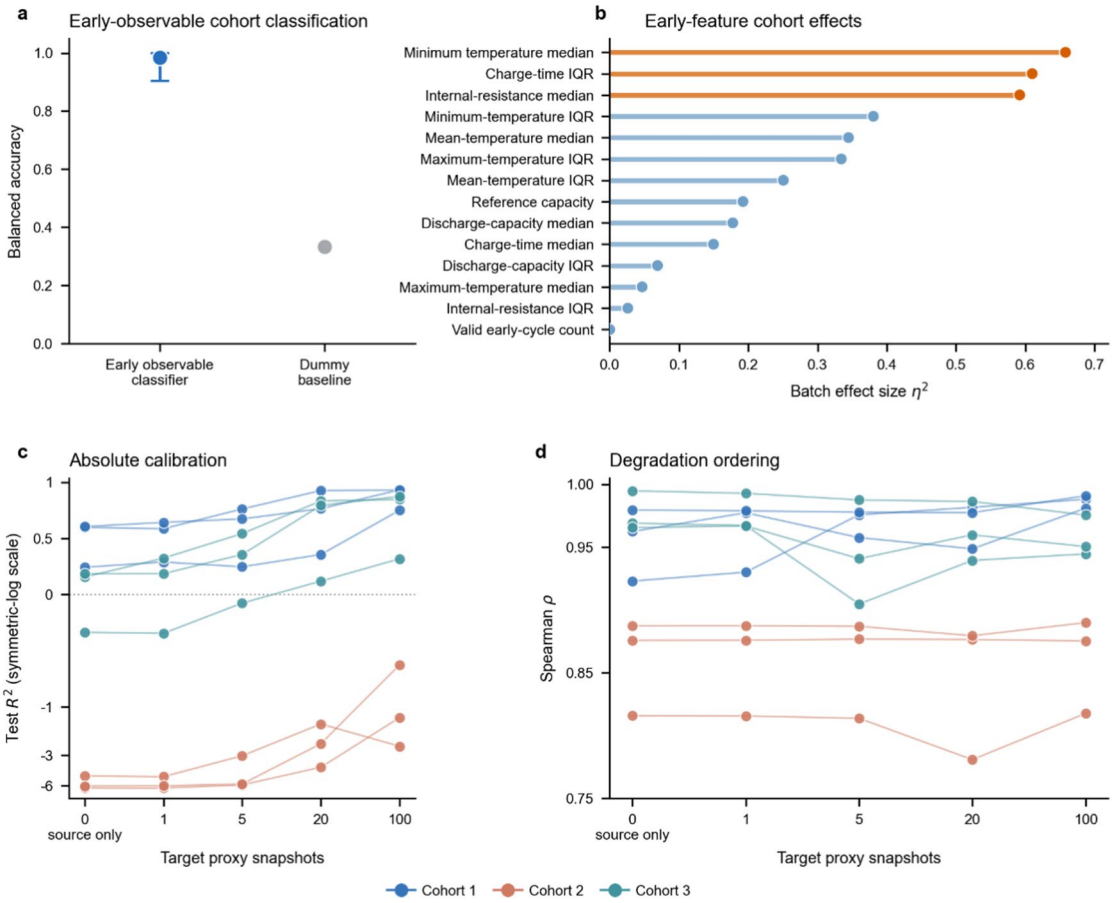



**Fig. 5** Source-cohort observability and barcode-disjoint cross-cohort calibration. (**a**) Balanced accuracy of the early-observable cohort classifier relative to a class-prior dummy baseline. The classifier point and error bar denote the mean and 2.5th–97.5th percentile range across repeated stratified fivefold cross-validation with ten repeats. (**b**) Cohort effect sizes ($\eta^2$) for the 14 early-observable features; orange markers identify the three largest effects. (**c**) Absolute calibration (test $R^2$) and (**d**) degradation ordering (Spearman ρ) as the number of barcode-disjoint target-proxy calibration snapshots increases. Colors denote the held-out target cohort, and lines connect matched cohort–seed evaluations across calibration budgets. The dotted line in (**c**) indicates $R^2 = 0$.

**Table 5** Measured electrical closure and member dynamics in the fixed 12S3P pack.

| Quantity | Scope | Members | Median \|error\| | 95th percentile | Identities | Transitions |
|---|---|---|---|---|---|---|
| Pack-current closure | All rows | 3 branches | 0.149 A | 0.600 A | — | — |
| Pack–branch voltage | All rows | 3 branches | 0.000 V | 0.0667 V | — | — |
| Branch–cell voltage | All rows | 36 cells | 0.047–0.048 V | 0.097 V | — | — |
| Member dynamics | Cycles 125–410 | 36 V / 72 T | — | — | 2 V / 6 T | 4 V / 37–72 T |

Note: Absolute residuals are summarized over 284,028 sampled rows. Member-identity counts refer to cycle-level discharge summaries.

low-cost independent transfer. Figure 5 summarizes early-observable source-cohort separability and the absolute-calibration and degradation-ordering responses to barcode-disjoint target calibration.

## Physical 12S3P electrical aggregation and member dynamics

The corrected external record contains one fixed 12S3P pack over 286 complete consecutive files: 217 WLTP cycles and 69 capacity checks from cycle 125 to 410. Measured electrical closure was strong. The median

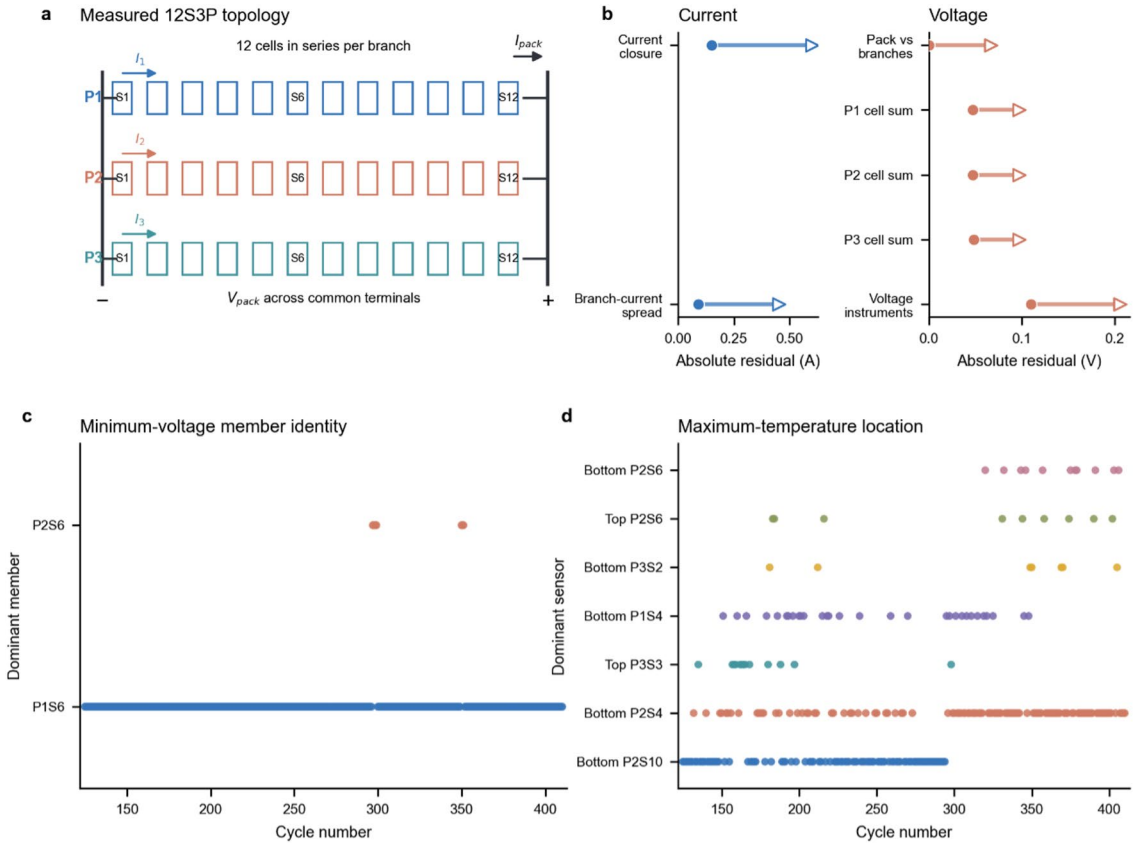



**Fig. 6** Physical reconstruction of one fixed 12S3P pack. (**a**) Measured topology; (**b**) voltage and current closure; (**c**) dominant minimum-voltage member; and (**d**) migration of the maximum-temperature location across capacity-check and WLTP cycles.

absolute residual between pack current and the sum of branch currents was 0.149 A (95th percentile, 0.600 A). Pack voltage minus mean branch voltage had median absolute residual 0 V (0.0667 V), and the three branch-voltage-minus-cell-sum residuals had medians of 0.047, 0.047 and 0.048 V (approximately 0.097 V at the 95th percentile).

The measured topology also exposed behaviours that an all-series tabular proxy cannot reproduce. Median branch-current spread was 0.088 A during capacity checks and 0.108 A during WLTP. P1S6 was the dominant minimum-voltage member in all 69 capacity-check cycles and in 212 of 217 WLTP cycles; P2S6 dominated the remaining five WLTP cycles, with four identity transitions. Thermal maxima were less stationary, involving six sensors and 37 transitions during capacity checks and five sensors and 72 transitions during WLTP. Member-aware monitoring is therefore physically relevant even when the voltage-limiting member is persistent.

The capacity checks do not supply an external RUL endpoint: measured capacity versus cycle number had Spearman correlation 0.016, and the discharge-cutoff protocol changed within the record. The earlier early-window capacity-prediction comparison is consequently withdrawn from the main evidence chain. The defensible external result is direct physical evidence for cell-to-branch-to-pack aggregation consistency, parallel-current imbalance and longitudinal member dynamics in one assembled pack.

These measurements connect the two proxy experiments to physical pack behaviour without conflating their targets. The longitudinal proxy tests whether aggregation features can follow fixed member compositions and remaining-life order; the 12S3P reconstruction verifies the electrical hierarchy and shows which member identities persist or migrate under coupled operation. Validation of absolute 16 S LEV RUL still requires multiple fixed physical packs with complete ageing endpoints.

## Discussion

The central result is that same-source fixed-member proxy trajectories retain sufficient longitudinal information for member-aggregated RUL learning across held-out compositions. This setting reflects the practical assembly of nominally comparable cells while preserving natural variation in capacity, resistance and degradation rate. High rank consistency across all three source cohorts shows that the aggregated observations contain a stable degradation-order signal. Absolute RUL is more demanding because the held-out endpoint must be represented by the augmented training compositions; this explains why strong Spearman correlation can coexist with poor or negative $R^2$. Mixed-source holdout and cross-cohort transfer extend this test beyond the primary assembly setting and expose additional distribution shift, but they are not required to justify the same-source use case.

Combinatorial member reuse broadens the range of training compositions available from a finite cell corpus. The matched sensitivity analysis showed that balancing barcode use and reducing composition overlap did not materially change rank consistency or absolute RUL performance, indicating that the longitudinal result was not driven by a small set of frequently selected cells. Because feasible overlap depends on source pool size, generated benchmarks should report composition uniqueness, pairwise overlap and per-barcode use rather than adopt a universal threshold. Barcode isolation between training and test members remains the essential safeguard against identity leakage.

The generated object is not limited to the RUL regression reported here. Compared with pack-only benchmarks that do not expose member-cell states, each trajectory retains member identity and cycle summaries alongside pack aggregates. The same structure can therefore support future pack-SOH, inconsistency-evolution, weak-member, abnormal-composition, and alternative-target studies. These are enabled uses of the data structure rather than additional tasks validated here.

The feature and target analyses clarify what the proxy model learns. Capacity-state descriptors carried most of the predictive signal, while resistance, temperature, and charge-time summaries provided smaller incremental contributions in this corpus. The minimum-member target is a conservative series-system label, but its disagreement with lower-decile and mean targets shows that it is not interchangeable with every physical retirement criterion[14–15,17–21,27–28]. The 12S3P reconstruction complements this statistical evidence by demonstrating current and voltage closure, parallel-current imbalance, and longitudinal member dynamics[9,29–31]. Multiple fixed physical LEV packs with complete ageing endpoints are still required to evaluate absolute deployment-level RUL.

The scope is also size-dependent. Member-wise construction is practical for compact LEV configurations with tens of cells, for which individual members and their dispersion can be represented directly. Packs containing hundreds or thousands of cells introduce module hierarchy, parallel current sharing, interconnect variation, and spatial thermal gradients. Extension to those systems should therefore use hierarchical, topology-aware module aggregation rather than merely increasing the member count.

The retained 148-cycle cell illustrates why an extreme short-lifetime member should not be removed solely because it degrades average model performance. In a physical pack, early degradation or abrupt cell-level anomalies would motivate BMS-assisted monitoring of member voltage, temperature and protection events, followed by decisions such as continued service, derating, repair, repurposing or recycling. The present independently tested-cell records demonstrate a short lifetime rather than an abrupt in-pack failure, and they contain no BMS event stream. Accordingly, this case supports a lifetime-coverage stress test, not a claim that the proxy model has already implemented physical fault diagnosis or intervention decisions.

## Conclusions

This study develops fixed-member longitudinal proxy-pack trajectories as a reproducible bridge from cell-ageing records to member-aware pack-level remaining-life assessment. In the same-source experiments, degradation ordering remained strong across all three public source cohorts, and tests within the augmented training endpoint

range achieved median $R^2$ of 0.909 and median Spearman correlation of 0.997. Absolute error increased when held-out lifetimes fell outside this range, identifying endpoint coverage as the principal calibration condition.

Supporting analyses showed that the main result was robust to training-composition reuse, while mixed-source and cross-cohort tests exposed the limits of domain transfer. The physical 12S3P reconstruction verified cell-to-branch-to-pack electrical closure and revealed both persistent voltage limitation and migrating thermal maxima. Retained member histories also enable future extension to other member-aware targets. The present flat member-wise formulation is intended for compact packs with tens of cells; larger systems require hierarchical module-level extensions. Together, these results position proxy-pack generation as an auditable benchmark for algorithm development from sparse pack-level evidence. Validation on multiple fixed physical LEV packs with complete ageing endpoints remains the next requirement for deployment-level RUL assessment.

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**Author contributions** Lu Zhu: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing. Wangwang Zheng: Resources, Data curation, Writing – review & editing. Fengmin Zhang: Supervision, Resources, Funding acquisition, Project administration, Writing – review & editing.

**Data availability** The raw cell-cycling data used to construct the proxy-pack benchmark are publicly available from the MATR data repository as the dataset associated with "Data-driven prediction of battery cycle life before capacity degradation" at https://data.matr.io/1/ (associated publication DOI: 10.1038/s41560-019-0356-8).The external 12S3P battery-pack dataset, "Lithium-Ion Battery Pack Cycling Dataset with CC-CV Charging and WLTP/Constant Discharge Profiles," is publicly available from the CORA Research Data Repository at https://doi.org/10.34810/data2395.The raw third-party datasets are not redistributed with this article and should be obtained directly from their original repositories. The processed proxy-pack tables, extracted feature files, and analysis scripts generated during the current study are available from the corresponding author upon reasonable request.

## Declarations

**Competing interests** The authors declare no competing interests.

**Declaration of generative AI and AI-assisted technologies in the writing process** During revision, the authors used a generative AI language tool to assist with language editing, structural comparison and code review. All scientific decisions, analyses, numerical results, references and final wording were checked and approved by the authors, who take full responsibility for the content of the manuscript.

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